Geometric Structure of GO Label Probes in Protein-Language-Model Embeddings: Spectral Concentration Is Explained by Label Co-Occurrence

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Abstract

Linear probes trained on protein-language-model embeddings are often read as revealing the directions along which a biological property is encoded. We test how far that reading is supported for 75 Gene Ontology Cellular Component (GO CC) labels represented in 480-dimensional ESM-2 embeddings of 9,895 proteins.

The stacked probe weight matrix $W \in \mathbb{R}^{75\times 480}$ is linearly independent (numerical rank 75) and has no near-collinear pair: the largest absolute cosine similarity between any two label directions is 0.4177, an angle of about 65°. Its singular-value spectrum is moderately concentrated: 90% of the squared spectral energy lies in 49 of the 75 directions, and the effective rank is 54.54.

We then ask whether that concentration is a property of the sequence representation. Three null models are compared, each preserving a different part of the problem. Random directions with matched row norms are less concentrated than the observed matrix (effective rank 60.91, $p < 0.001$), and weights fitted to column-shuffled labels are also less concentrated once row norms are matched (57.75, $p = 0.005$). However, a row-permutation null that preserves the co-occurrence structure of the labels but breaks its association with the proteins reproduces the observed spectrum almost exactly (54.32, $p = 0.73$).

The concentration is therefore inherited from the statistical structure of the GO labels rather than from protein-specific information extracted by the probe. This is also what the geometry of the problem predicts: with 75 labels in 480 dimensions there is no capacity pressure that would call for superposition in the first place. Pairwise weight similarity still tracks label co-annotation after controlling for co-occurrence (partial rank correlation 0.4889), so the geometry carries semantic information, but it is not independent evidence of feature superposition.

1 Introduction

Pretrained protein language models map amino-acid sequences to continuous vectors. If such a vector contains information about subcellular localization, a linear classifier can recover part of it, and its weight vector is often interpreted as the direction along which the model represents the corresponding property.

This interpretation becomes more interesting when several labels appear to share directions. Elhage et al. [1] describe *superposition*: a network may encode more features than it has dimensions, allowing features to share directions at the cost of interference. A natural question is whether the weight vectors of many GO labels, trained independently on the same embeddings, display a similar phenomenon.

That analogy needs care. A probe is a supervised read-out fitted on frozen features, not a description of the model's internal computation, and a shared direction between two probes can arise for entirely mundane reasons: the labels are correlated, the embedding covariance is anisotropic, or the training procedure regularizes in a particular way. Evidence for superposition therefore has to distinguish geometric structure that follows from the label distribution from structure that the representation itself contributes.

This report asks four questions about the weight matrix W of 75 GO CC probes:

Q1. Are any two weight vectors exactly collinear?

Q2. Do any three weight vectors span a subspace of dimension less than three?

Q3. Is any subset of M weight vectors rank-deficient?

Q4. Even without exact dependence, is the spectrum of W concentrated beyond what a suitable null model produces?

Questions Q1–Q3 have clean linear-algebraic answers. Q4 is the substantive one, because approximate shared structure is what machine-learning models actually exhibit at finite numerical precision, and because a positive answer only becomes meaningful once the null model is specified.

Our contribution is threefold. First, we answer Q1–Q3 with quantitative margins rather than a rank test alone. Second, we quantify the spectral concentration and show that it is moderate, not a collapse onto a few directions. Third, and most importantly, we compare the observed spectrum against a ladder of null models that preserve progressively more of the label structure, and show that the concentration is reproduced when the labels' co-occurrence pattern is preserved but their association with the proteins is broken.

That last result changes the interpretation a concentrated spectrum would otherwise invite. The concentration is real, but it is a property of the GO label distribution as seen through a linear read-out, not evidence that ESM-2 spreads biological features across shared directions.

2 Data and Model

2.1 Dataset

We download reviewed (Swiss-Prot) entries from the UniProtKB REST API and keep those with at least one GO Cellular Component annotation [3, 4]. The download is capped at 10,000 records, which keeps the experiment reproducible on a laptop; the resulting dataset is a prefix of reviewed Swiss-Prot rather than the full database. Entries without an accession, sequence, or GO CC annotation are dropped, and the 75 GO CC terms carried by at least 100 proteins are kept. Every sequence is embedded with ESM-2 `esm2_t12_35M_UR50D` [2], using the mean of the layer-12 states over the residues (excluding the BOS and EOS tokens) as the protein vector. Table 1 summarises the dataset.

Table 1: Dataset used in this study.

Item	Value
Protein records	9,895
Embedding dimension	480
Selected GO CC labels	75
Distinct GO CC terms in the export	1,779
Label matrix Y	9895×75
Positive label assignments	36,188
Mean labels per protein	3.6572
Proteins with at least one selected label	9,786
Embedding matrix X	9895×480

2.2 Linear probe

For protein n with embedding $x_n \in \mathbb{R}^{480}$ and label vector $y_n \in \{0,1\}^{75}$, a one-vs-rest logistic regression is fitted with balanced class weights and L_2 regularization $C = 1$:

$$p(y_{ni} = 1 \mid x_n) = \sigma(w_i^\top x_n + b_i), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{1}$$

Stacking the 75 coefficient vectors gives

$$W = \begin{bmatrix} w_1^\top \\ \vdots \\ w_{75}^\top \end{bmatrix} \in \mathbb{R}^{75 \times 480}, \tag{2}$$

whose rows are the objects of study. The embeddings are standardised inside the pipeline, so W is expressed in the standardized coordinate system.

A single 80/20 split with a fixed seed (42) gives 7,916 training and 1,979 test proteins. On the test split the mean ROC-AUC is 0.7841 and the mean average precision is 0.2226, averaged over the labels that contain both classes in the fold. The probe is therefore informative, but the performance metrics say nothing about the geometry analysed below; they are reported to establish that W is a meaningful read-out rather than an artefact of an untrained model.

3 Methods

3.1 Exact linear dependence (Q1–Q3)

For two nonzero weight vectors, the cosine similarity is

$$c_{ij} = \frac{w_i^\top w_j}{\|w_i\|_2 \, \|w_j\|_2}. \tag{3}$$

The vectors are collinear exactly when $|c_{ij}| = 1$, equivalently when $\operatorname{rank}([w_i^\top; w_j^\top]) < 2$. The absolute value is needed because antiparallel vectors are also linearly dependent.

A rank test alone says little about how far a matrix is from dependence. For two unit vectors with cosine c, the singular values of the 2×480 stack are $\sigma_1^2 = 2(1 + |c|)$ and $\sigma_2^2 = 2(1 - |c|)$, so

$$\frac{\sigma_2}{\sigma_1} = \sqrt{\frac{1-|c|}{1+|c|}}. \tag{4}$$

This ratio turns a cosine into a collinearity margin and is reported together with the rank. Numerical ranks use the standard tolerance $\max(m, n)\,\epsilon\,\sigma_1$ with ϵ the machine epsilon.

Q2 and Q3 are logical consequences of Q1 for a full-row-rank matrix: any subset of linearly independent rows is also independent. We nevertheless verify them numerically on random row subsets, which also serves as a check on the tolerance.

3.2 Spectral concentration (Q4)

The singular value decomposition of W is

$$W = U\Sigma V^\top, \qquad \sigma_1 \geq \cdots \geq \sigma_{75} \geq 0. \tag{5}$$

The squared singular values measure the energy along each orthogonal direction. We report the cumulative energy

$$E(k) = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{75} \sigma_i^2}, \tag{6}$$

the number of directions needed for 90% and 95% of the energy (Rank90, Rank95), and the effective rank

$$r_{\text{eff}} = \exp\left(-\sum_{i:\,p_i>0} p_i \log p_i\right), \qquad p_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2}. \tag{7}$$

Effective rank is the continuous analogue of algebraic rank: it equals 75 for a flat spectrum and approaches 1 for a rank-one matrix.

3.3 Null models

A concentration statistic is only interpretable relative to a null. We use three, chosen so that each preserves a different part of the original problem (Table 2).

Table 2: Null models. Each row states what is randomized and what is held fixed.

Null	Randomized	Preserved
Isotropic directions	the direction of every row, redrawn from a Gaussian	row norms, matrix shape
Label shuffle	each label column, independently	number of positives per label, training embeddings
Row permutation	the pairing between proteins and label vectors, by permuting the rows of Y as a block	label co-occurrence structure, label prevalence, training embeddings

In every case the random weight matrix is rescaled so that its row norms match those of the observed W. Row norms matter because the spectrum of a matrix depends on its magnitude profile as well as on the angles between rows; without matching, a null can land on either side of

the observation for reasons unrelated to direction structure. The label-shuffle null is therefore evaluated both raw and matched, and the contrast between the two is itself informative.

For a statistic T where lower values mean more concentration, the empirical lower-tail p-value over B draws is

$$p = \frac{1 + \#\{T_b \leq T_{\text{obs}}\}}{1 + B}. \tag{8}$$

All nulls use fixed seeds (42 for the isotropic and label-shuffle nulls, 2026 for the row-permutation null) and are computed on the same training embeddings and the same probe hyper-parameters as the observed matrix.

3.4 Association with label structure

If the geometry follows the labels, labels that are annotated together should receive similar probe directions. For each of the $\binom{75}{2} = 2{,}775$ unordered label pairs we compute the label correlation r_{ij} (Pearson correlation of the two binary columns), the co-occurrence count q_{ij}, and the weight cosine c_{ij}.

The monotone association between c_{ij} and r_{ij} is measured with Spearman's rank correlation. Because both quantities grow with label frequency, we also compute a residualized version: c_{ij} and r_{ij} are rank-transformed, each is regressed on an intercept and the rank of $\log(1 + q_{ij})$, and the correlation of the residuals is reported. This controls for co-occurrence rank only; it does not control for ontology structure or label prevalence.

4 Results

4.1 Exact geometry: no dependence, with a wide margin

The weight matrix has shape 75×480 and numerical rank 75 (tolerance 4.5×10^{-12}). Every random subset of rows is full rank, including samples of all 75 rows (Table 3).

Table 3: Rank of random row subsets (100 samples per subset size).

M	Mean rank	Rank-deficient subsets
2	2.00	0/100
3	3.00	0/100
5	5.00	0/100
10	10.00	0/100
20	20.00	0/100
40	40.00	0/100
75	75.00	0/100

The margins are wide. The most similar pair of weight vectors is `GO:0005789` (endoplasmic reticulum membrane) and `GO:0005783` (endoplasmic reticulum), with $|c| = 0.4177$; the corresponding angle is 65.3° and $\sigma_2/\sigma_1 = 0.641$. The most anti-aligned pair reaches $c = -0.161$, and the mean absolute cosine over all pairs is 0.063. No triple or larger subset is degenerate.

The negative answer to Q1–Q3 is therefore robust to the choice of tolerance: it is not a borderline numerical call.

4.2 Spectral concentration

The spectrum decays gradually (Figure 1). The leading singular value is $\sigma_1 = 41.96$ and accounts for 7.6% of the total energy; the first 20 directions account for 54.7%. The 90% level is reached at $k = 49$ and the 95% level at $k = 57$ (Figure 2, Table 4). The effective rank is 54.54 out of a possible 75.

Table 4: Spectral summary of W.

Quantity	Value
σ_1	41.96
Rank90	49
Rank95	57
Effective rank	54.54
Maximum attainable rank	75
Mean absolute cosine between rows	0.063

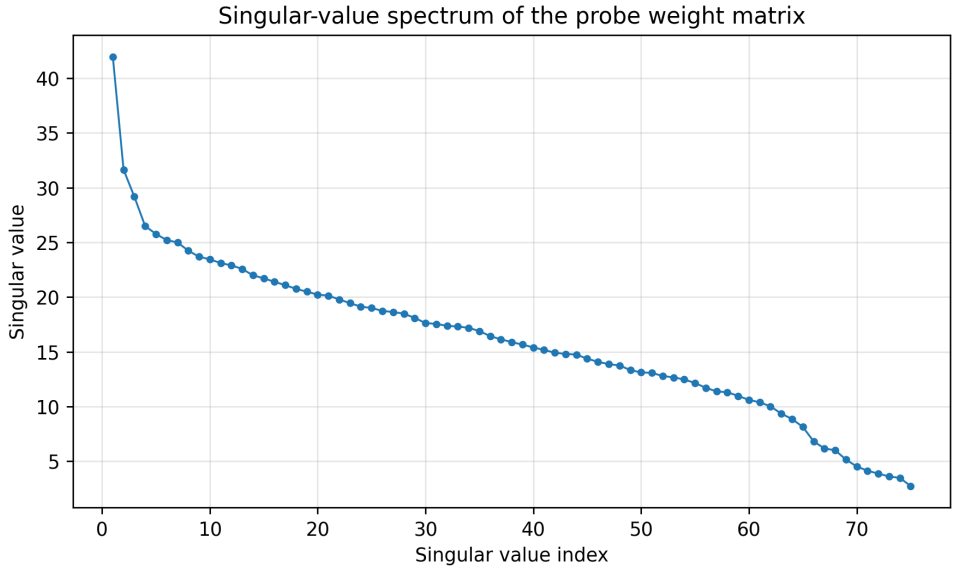


Figure 1: Singular-value spectrum of the probe weight matrix. The decay is continuous: there is no gap separating a small leading subspace from the rest.

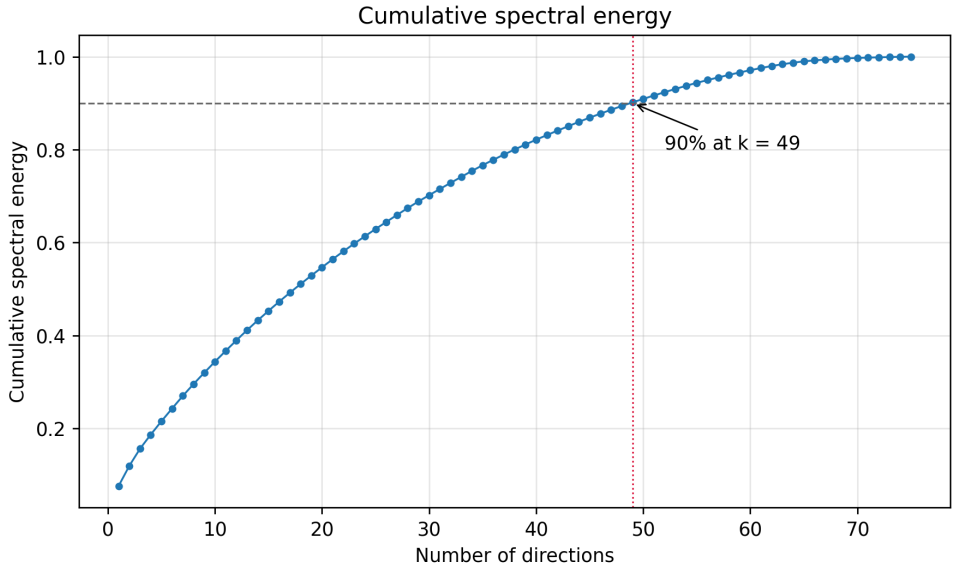


Figure 2: Cumulative spectral energy. The dotted line marks the 90% level, reached at 49 of the 75 directions.

The concentration is moderate. A matrix that genuinely occupied a 5- or 10-dimensional subspace would reach 90% of its energy within a handful of directions and show a sharp spectral gap; here the curve is close to linear through the middle of the spectrum, and two thirds of the directions are still needed to reach 90%.

4.3 The concentration is reproduced by label structure alone

Figure 3 and Table 5 compare the observed spectrum with the null models of Section 3.3.

Table 5: Null models compared with the observed spectrum. Intervals are 95% empirical intervals; p-values are lower-tail. Lower values mean more concentration.

Null	B	Effective rank	p	Rank90	p
Observed W	–	54.54	–	49	–
Isotropic directions	1000	60.91 [60.66, 61.14]	< 0.001	51.96 [51, 52]	< 0.001
Label shuffle, raw	200	52.51 [52.04, 53.03]	1.000	44.24 [44, 45]	1.000
Label shuffle, matched	200	57.75 [57.36, 58.16]	0.005	49.65 [49, 50]	0.358
Row permutation	100	54.32 [53.57, 54.98]	0.733	47.98 [48, 48]	1.000

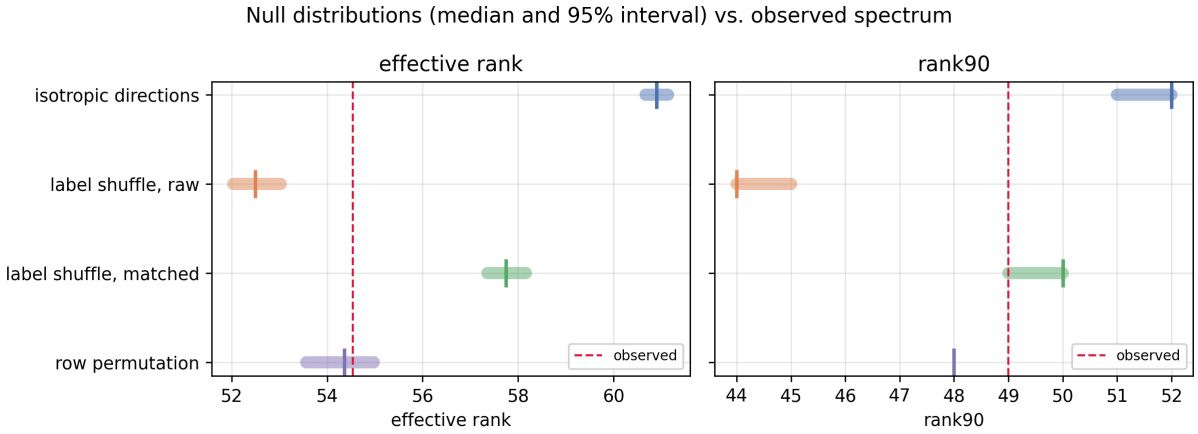


Figure 3: Null distributions (median and 95% interval) against the observed effective rank and Rank90. The row-permutation null is the only one that coincides with the observation.

Read from top to bottom, the table is a ladder. Random directions of the right magnitude are less concentrated than W ($p < 0.001$), so the weights are not noise. Shuffling labels destroys the co-occurrence pattern and moves the null further from the observation. Keeping the co-occurrence pattern while breaking its association with the proteins brings the null to 54.32, against an observed 54.54 ($p = 0.73$) and a Rank90 of 47.98 against 49.

The last row is decisive for the interpretation. If the observed spectrum is what one obtains by fitting the same probe to label vectors that carry the correct co-occurrence structure but are attached to the wrong proteins, then the spectrum is a consequence of the label distribution and the regression, not of protein-specific information in the embeddings. The sequence representation still carries predictive signal (mean ROC-AUC 0.784), but the concentration of W is not evidence of additional geometric structure contributed by that representation.

The raw label-shuffle row is a reminder that null definitions matter: without matching row norms it is *more* concentrated than the observation ($p = 1.0$ in the lower tail). A study that quoted only that row would reach the opposite conclusion. Every null in this report is therefore reported with its matching rule and its direction of comparison.

4.4 Weight similarity tracks label co-annotation

Figure 4 shows the association between label correlation and weight-vector cosine over all 2,775 pairs, and Figure 5 shows the full cosine matrix.

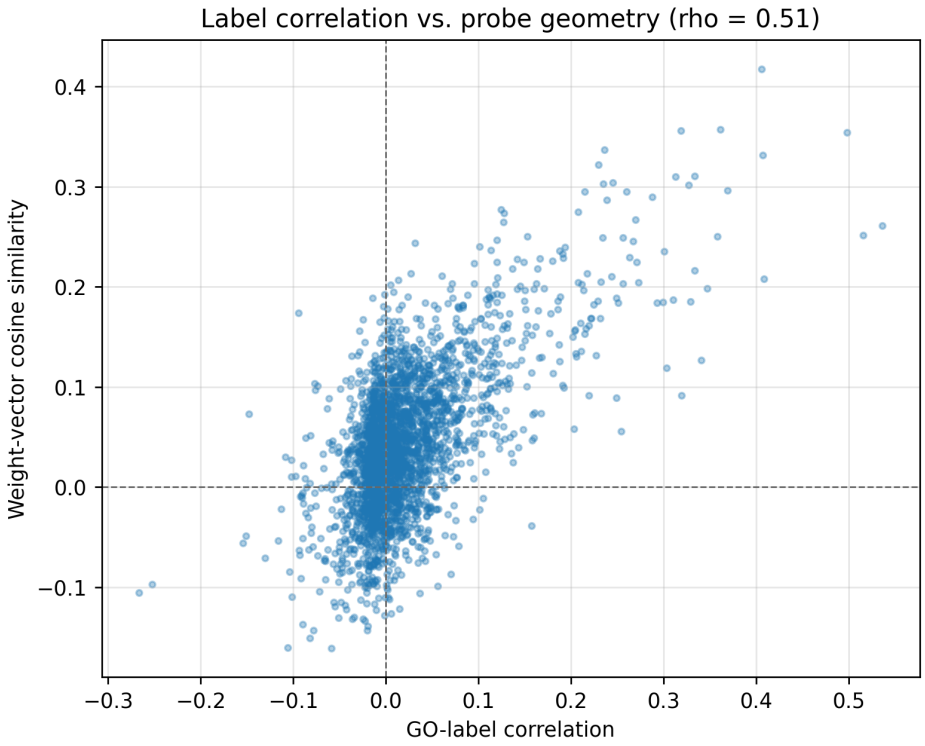


Figure 4: Label correlation against probe weight-vector cosine similarity, one point per unordered label pair.

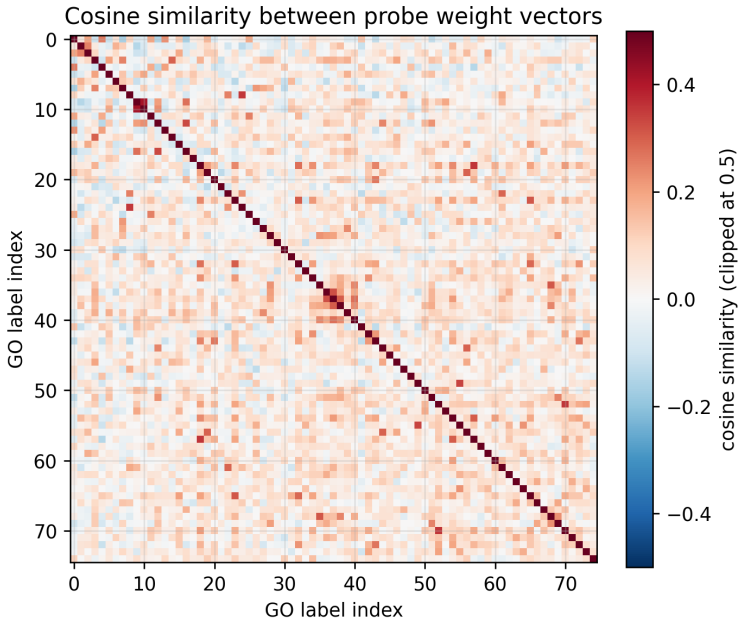


Figure 5: Cosine similarity between the 75 probe weight vectors (colour scale clipped at ±0.5). Off-diagonal similarities are small but structured: a few pairs and a small cluster of endoplasmic-reticulum terms reach 0.2-0.4, and no pair approaches 1.

Spearman's rank correlation is $\rho = 0.5074$ (nominal $p = 1.7 \times 10^{-181}$). After residualizing both variables on the rank of $\log(1+q_{ij})$, the partial rank correlation is 0.4889 (nominal $p = 1.0 \times 10^{-166}$).

The association therefore survives the co-occurrence control and is not an artefact of frequency.

The strongest pairs are biologically adjacent: endoplasmic reticulum membrane versus endoplasmic reticulum ($c = 0.418$, $r = 0.406$, 332 shared proteins), presynaptic versus postsynaptic membrane, and glutamatergic synapse versus postsynapse. The anti-aligned end pairs the nucleus with peripheral compartments such as the extracellular exosome ($c = -0.161$, $r = -0.059$). Among the 1,218 pairs with non-positive label correlation the largest cosine is still 0.192, so the relationship is a strong trend rather than a deterministic rule.

Taken with Section 4.3, these results describe a consistent picture: the probe assigns similar directions to labels that co-occur, and the resulting correlation among rows is what produces the moderately concentrated spectrum.

5 Discussion

5.1 What the geometry shows

Three statements are supported by the data. The 75 probe directions are linearly independent with a wide margin, so there is no exact redundancy. Their spectrum is moderately concentrated, with an effective rank of 54.54 and 49 directions needed for 90% of the energy. And the concentration is reproduced when the co-occurrence structure of the labels is preserved but its association with the proteins is destroyed.

The third statement is the informative one. It says that the geometry of W is a read-out of the label distribution: terms such as "endoplasmic reticulum" and "endoplasmic reticulum membrane" are almost always co-annotated, so their probes converge on similar directions, and the set of 75 directions is more correlated than random. No property of ESM-2 is needed to explain the spectrum beyond the fact that it supports a linear read-out at all.

5.2 Why this is not evidence of superposition

The superposition hypothesis [1] is about capacity: a network encodes more features than it has dimensions and accepts interference as the price. Nothing analogous applies here. The probe maps 480-dimensional embeddings to 75 labels, so its 75 directions live in a space more than six times larger than the number of features; there is no pressure to share them. The regime here is the opposite of capacity pressure: the probe has far more dimensions available than labels to represent.

This is also why the observed effective rank of 54.54 is expected rather than surprising. Correlated labels produce correlated rows, and correlated rows produce a spectrum that falls short of full rank. The number 54.54 is a statement about how redundant the GO CC annotation is, not about how the model stores features.

Establishing superposition would require different evidence: features that outnumber dimensions, interference that can be measured behaviourally, causal interventions on individual directions, or sparse dictionary recovery that recovers more features than the representation dimension. None of these is available from a set of independent logistic probes.

5.3 Limitations

1. **Data scope.** The download is capped at 10,000 reviewed records, and only 75 GO CC terms pass the 100-protein threshold. The dataset is a prefix of reviewed Swiss-Prot, so the absolute numbers are not comparable across papers.

2. **Single split, no homology control.** A single 80/20 split is used, without homology-aware

partitioning. Proteins with near-identical sequences can appear in both halves, which would inflate the ROC-AUC. The geometry analysis does not depend on the test split, but the reported performance does.

3. **One representation.** Only layer 12 of `esm2_t12_35M` is probed, with mean pooling. Other layers, pooling schemes, regularisation strengths, and larger checkpoints may produce different spectra.

4. **Dependent label pairs.** The 2,775 pairs share labels, so the nominal p-values of Section 4.4 are descriptive rather than inferential. Label-level or protein-level resampling would be required for formal inference.

5. **Null-model scope.** Each null preserves a different subset of the problem, and none preserves everything. The row-permutation null destroys the protein-label pairing entirely, so it describes what labels alone can produce; it does not prove that labels are the only cause of the observed spectrum.

6. **Ontology structure.** GO terms are related by `is_a` and `part_of`, and no propagation over the ontology is applied. Part of the co-occurrence structure is therefore ontological rather than empirical.

5.4 Future work

1. **Capacity-constrained probes.** Represent the same labels in a space smaller than 75 dimensions, so that directions must be shared, and check whether prediction degrades in the pattern superposition predicts. This is the test that most directly addresses the hypothesis.

2. **Stability of the leading subspace.** Refit W on repeated training subsets, regularisation strengths, and random seeds, and compare the leading k-dimensional subspaces. For orthonormal bases Q_1, Q_2 of two subspaces, the similarity

$$S = \frac{\|Q_1^\top Q_2\|_F^2}{k} \tag{9}$$

   lies in $[0, 1]$. A subspace that is stable across seeds reflects the task; one that moves reflects optimisation.

3. **Layers and pooling.** Repeat the analysis for other ESM-2 layers, attention-weighted pooling, and larger checkpoints. If the concentration is inherited from the labels, it should be largely invariant to these choices.

4. **Homology-aware evaluation.** Partition train and test by sequence similarity rather than at random, and report per-label metrics with confidence intervals from repeated splits.

5. **Ontology-aware labels.** Propagate annotations over the GO graph, which would make the dependence between labels explicit and test whether the residual concentration survives once ontology structure is modelled directly.

6 Reproducibility

All results are produced by two notebooks, both published in the project repository together with the cached data tables and the source of this report [5]:

- `Protein.ipynb` builds the dataset: it downloads the UniProt records, filters and parses the GO annotations, selects the labels, and computes the ESM-2 embeddings.

- `protein_go_geometry.ipynb` contains every analysis reported here: the probe, the rank checks, the spectrum, the three null-model families, the label-structure analysis, and the figures. It starts from the cached arrays in `protein_dataset/` and runs in about 11 minutes on a CPU, dominated by the 300 probe refits of the permutation nulls.

The analysis notebook writes `results.json`, which holds the main statistics quoted in this report, and the five figures in `figures/`. Re-running it regenerates both, so the manuscript can be checked against the code by comparing `results.json` with the tables above. The probe weight matrix is saved to `protein_dataset/probe_weights.npy` so that later sections can be re-run without refitting.

7 Conclusion

For 75 GO Cellular Component labels represented in 480-dimensional ESM-2 embeddings, the probe weight matrix is linearly independent, has no near-collinear pair, and shows moderate spectral concentration (effective rank 54.54, Rank90 49).

That concentration is not evidence that the representation stores features in shared directions. A null model that preserves the co-occurrence structure of the labels but breaks its association with the proteins reproduces the observed spectrum almost exactly (54.32, $p = 0.73$), while nulls that destroy the label structure do not. The geometry of the probe is a read-out of how redundant the GO annotations are, and with 75 labels in 480 dimensions there is no capacity pressure that would make superposition necessary.

Weight-vector similarity nevertheless tracks label co-annotation (Spearman 0.5074, partial 0.4889), so the probe's directions remain semantically interpretable at the level of pairs. The distinction matters: the directions are meaningful, but their global geometry is inherited rather than learned.

Author contributions

R.M. designed the analysis, implemented the code, and wrote the manuscript. G.L. contributed to the project.

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